

PINGCHENG JIAN

PhD student, Electrical and Computer Engineering, Duke University

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EDUCATION

Duke University

Ph.D. in Electrical and Computer Engineering

Sep. 2021 - present

Advisors: [Michael Zavlanos](#), [Boyuan Chen](#)

Tsinghua University

B.E. in Automation

Transfer major from Mechanical Engineering (2016-2018)
to Automation (2018-2021)

Aug. 2016 - Jul. 2021

Advisors: [Huaping Liu](#), [Mingguo Zhao](#)

RESEARCH INTERESTS

My area of research is robot learning, and my previous publications cover the 3 topics: 1) applying foundation models for the preference feedback in reinforcement learning, 2) modularity for robot learning, and 3) adversarial learning for robust control.

PUBLICATIONS

(* equal contributions, ** alphabetical order. Up-to-date list can be found in my Google Scholar page.)

Journal Papers

1. **[TMLR'24]** [Pingcheng Jian](#), Easop Lee, Zachary I. Bell, Michael M. Zavlanos, Boyuan Chen, "Perception Stitching: Zero-Shot Perception Encoder Transfer for Visuomotor Robot Policies", *Transactions on Machine Learning Research (TMLR)*, 2024. [\[openreview\]](#) [\[website\]](#) [\[video\]](#) [\[arxiv\]](#)
2. **[TMLR in submission]** [Pingcheng Jian](#), Xiao Wei, Yanbailui Liu, Samuel A. Moore, Michael M. Zavlanos, Boyuan Chen, "LAPP: Large Language Model Feedback for Preference-Driven Reinforcement Learning", *In submission to Transactions on Machine Learning Research (TMLR)*, 2025. [\[video\]](#) [\[arxiv\]](#)

Conference Papers

3. **[CoRL'23]** [Pingcheng Jian](#), Easop Lee, Zachary Bell, Michael M. Zavlanos, Boyuan Chen, "Policy Stitching: Learning Transferable Robot Policies", *Conference on Robot Learning (CoRL)*, 2023. [\[website\]](#) [\[CoRL talk\]](#) [\[video\]](#) [\[arXiv\]](#)
4. **[ICRA'21]** [Pingcheng Jian*](#), Chao Yang*, Di Guo, Huaping Liu, Fuchun Sun, "Adversarial Skill Learning for Robust Manipulation", *International Conference on Robotics and Automation (ICRA)*, 2021. [\[pdf\]](#) [\[video\]](#)
5. **[L4DC'23]** Reza Khodayi-mehr, [Pingcheng Jian](#), Michael M. Zavlanos, "Physics-Guided Active Learning of Environmental Flow Fields", *Learning for Dynamics and Control (L4DC)*, 2023. [\[pdf\]](#)

Workshop Papers

6. **[ICML-W'21]** Minghao Zhang*, [Pingcheng Jian*](#), Yi Wu, Huazhe Xu, Xiaolong Wang, "DAIR: Disentangled Attention Intrinsic Regularization for Safe and Efficient Bimanual Manipulation", *International Conference on Machine Learning (ICML) Workshop: Reinforcement Learning for Real Life*, 2021. [\[website\]](#)

PATENT

US Patent (pending): Large Language Model-Assisted Preference Prediction for Reinforcement Learning (Application No. 63/751,383, filed on January 30, 2025; Attorney Docket No.: DU8724PROV)

中国发明专利: 一种物流仓库中物品的装箱方案生成方法及系统 (授权公告号: CN110443549B, 申请号: 2019106808700)

PROJECTS

Quadruped Spider Robot, course project of *CS590 - Robot Studio* at Duke University

Project Video: [\[video\]](#)

I built all the hardware and software of this spider robot from scratch. I drew the parts of this robot with SOLIDWORKS and then printed them with 3D printer. The controller of this robot is a Raspberry Pi, and the control algorithm is written in Python. Then I generated the .urdf files of this robot and built the simulator of this robot with PyBullet.

ACADEMIC SERVICE

Reviewer:

Conference on Robot Learning (CoRL), 2024

Conference on Robot Learning (CoRL), 2025

TEACHING EXPERIENCES

Teaching Assistant at Duke University

[ECE 382L/ME 344L](#) - Control of Dynamic Systems [\[lecture\]](#) [\[recitation\]](#)

[ECE 689/COMPSCI 676](#) - Advanced Topics in Deep Learning

ADVISING AND MENTORSHIP

Easop Lee, Duke Undergraduate Student in ECE (2022). Next: Ph.D. at Duke ECE

Xiao Wei, Duke CS Master (2024 - present)

Yongjae Lee, Duke ME Master (2024 - present)

SKILLS

Programming Languages

Python, Matlab, C, C++, C#, HTML, Assembly Language

Machine Learning

Pytorch, Reinforcement Learning, Supervised Learning

Robotic Software

PyBullet, MuJoCo, Isaac Gym

Robotic Hardware

SOLIDWORKS, 3D printing